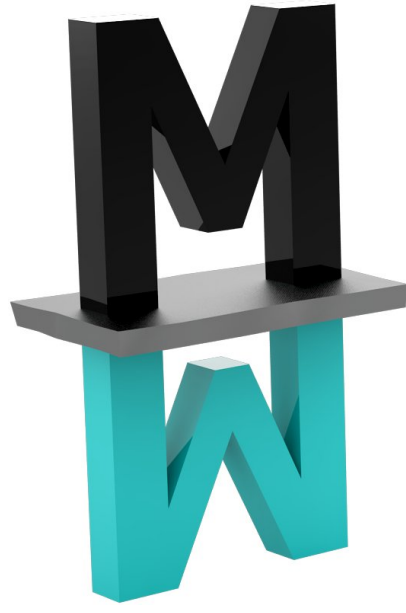




1. Introduction

- ▶ Intentionally and controllably designed **non-trivial material architectures** can bring behaviour **outperforming homogeneous bulk** materials.
- ▶ **Superlattices** represent perspective route for protective coatings. These, however, often require outstanding mechanical properties.
- ▶ For a knowledge-driven development of new systems, reliable and **predictive modelling** is required.
- ▶ *Ab initio* methods are accurate but computationally expensive; **continuum mechanics** treatment is affordable, but characterisation of interface and/or surface properties is required.

↔ Method of Grimsditch and Nizzoli [1, 2]

2. The G&N method

Linear-elasticity continuum model-based on elasticity tensor, molar ratio and crystallographic orientation of each individual phase.

Limitations:

1. **Neglects** any heterogeneity introduced by the **layer interface**.
2. **Neglects** the **bi-layer period**.
3. The input C_{ij} s correspond to equilibrium **lattice parameters** of each phases, which can differ from the in-plane lattice parameter of the superlattice.

G&N for 2 layers: $\hat{C}_{ij} = \chi(\hat{C}_1, f_1, \hat{C}_2, f_2)$

$$1. \hat{P}_i = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ C_{31} & C_{32} & C_{33} & C_{34} & C_{35} & C_{36} \\ C_{41} & C_{42} & C_{43} & C_{44} & C_{45} & C_{46} \\ C_{51} & C_{52} & C_{53} & C_{54} & C_{55} & C_{56} \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

$$2. \hat{M}_i = \hat{P}_i^{-1} \hat{P}_1, i = 1, 2$$

$$3. \hat{M} = \sum_{i=1}^2 f_i \hat{M}_i$$

$$4. C_{ij}(\text{bi-layer}) = (\sum_{i=1}^2 f_i \hat{C}_i \hat{M}_i) \hat{M}^{-1}$$

Can be generalised for N layers.

3. Conclusions

- ▶ An explicit *ab initio* elasticity calculations of nitride **superlattices and a grain boundary** in Ni_3Al intermetallic alloys.
- ▶ Continuum linear elasticity approach as formulated by **Grimsditch and Nizzoli** was applied.
- ▶ CrN/AlN superlattice yielded **excellent agreement** between explicit *ab initio* elastic tensor and its approximation by the G&N method.
- ▶ **Qualitative agreement** obtained also for majority of MoN/TaN systems.
- ▶ The G&N method **fails** for the case of Σ_5 GB in Ni_3Al .
- ▶ Reasons are:
 - omission of **interface-induced modifications** of atomic/electronic structure,
 - inability to reflect different **bi-layer periods**.

References

- [1] M. Grimsditch, Phys. Rev. B Condens. Matter **31**, 6818 (1985).
[2] M. Grimsditch and F. Nizzoli, Phys. Rev. B Condens. Matter **33**, 5891 (1986)

Acknowledgements

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